

NETWORK REQUEST MANAGEMENT SYSTEM

1. INTRODUCTION

This document presents the User Acceptance Testing (UAT) report for the Network Request Management System project developed on the ServiceNow platform. The project was built as part of an individual Skill Wallet project to automate the end-to-end lifecycle of network access requests in an organization.

The testing covers all key functionalities including Service Catalog submission, approval workflow, task fulfilment, role-based access control, and email notifications. All test cases were executed on the ServiceNow Personal Developer Instance (PDI) and the results are documented in this report.

1.1 PROJECT OVERVIEW

Project Name : Network Request Management System

Project Version : 1.0

Project Type : Individual Project

Platform : ServiceNow Personal Developer Instance

Testing Period : 22 June 2026 to 28 June 2026

Prepared By : NITHISHA B

Project Description:

The Network Request Management System project is built on the ServiceNow platform to streamline and automate the end-to-end lifecycle of network access requests within an organization. Traditionally, employees request network access (VPN, Wi-Fi, LAN, Firewall, or IP Address) through manual emails or forms, resulting in delays, lack of visibility, and inconsistent tracking.

This project addresses those challenges by implementing a standardized Service Catalog item, an automated approval workflow using Flow Designer, auto-created fulfilment tasks for the Network Team, role-based access control for exception handling, and email notifications at every stage — all accessible through the ServiceNow Service Portal.

Key Components Built:

- ✓ Service Catalog Item with 5 variables (Request Type, Justification, Attachment, Duration, Urgency)
- ✓ Flow Designer Workflow (Manager Approval + Create Catalog Task)
- ✓ ACL / Role-based access control — restricts approval actions to Manager role only
- ✓ Email notifications on approval, rejection, and task completion
- ✓ Update Set exported and deployed to another PDI instance

1.2 PURPOSE

The purpose of this User Acceptance Testing (UAT) document is to:

- I. Verify that the Network Request Management System functions correctly as per the defined requirements and user stories
- II. Ensure that the Network Request catalog item captures all required employee request details and triggers the workflow correctly
- III. Validate that the Flow Designer workflow routes approval requests to the correct Manager automatically
- IV. Confirm that fulfilment tasks (sc_task) are auto-created and assigned to the Network Team upon approval
- V. Test that email notifications are sent to the requester at every stage — submission, approval, rejection, and task completion

- VI. Verify that access control correctly restricts approval actions to authorized Manager roles only
- VII. Confirm that the complete project is successfully exported as an Update Set XML and deployed to another ServiceNow PDI instance without any missing elements
- VIII. Ensure the overall solution meets the business objective of reducing manual effort, improving visibility, and enforcing IT governance compliance

2. IDEATION PHASE

2.1 PROBLEM STATEMENT

Date	22 June 2026
Team ID	
Project Name	Network Request Management System
Maximum Marks	2 Marks

Customer Problem Statement:

Create a problem statement to understand your customer’s point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you’ll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

I am	Describe the customer and their attributes here
I'm trying to	List the thing they are trying to achieve here
but	Describe the problems or barriers that get in the way here
because	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions the result from experiencing the problems or barriers

Example:

I am	I'm trying to	but	because	which makes me feel
a traveler	book flights on my phone	it takes a long time	The website is not responsive and doesn't have a mobile version	Frustrated

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	An Employee who needs VPN / Wi-Fi / LAN / Firewall or IP access	Submit a network service request and get it approved quickly	The process is manual, email-based and has no tracking system	There is no centralized platform to raise and manage network requests	Frustrated and uncertain about the status of my request
PS-2	A Network Engineer assigned to process network access requests	Complete and close network tasks assigned to my team efficiently	Requests arrive informally with no structured task or priority system	There is no automated workflow to route approved requests to the right team	Overwhelmed and unable to track or prioritise pending work

2.2 EMPATHY MAP CANVAS

Date	22 June 2026
Team ID	
Project Name	Network Request Management System
Maximum Marks	4 Marks

Empathy Map Canvas:

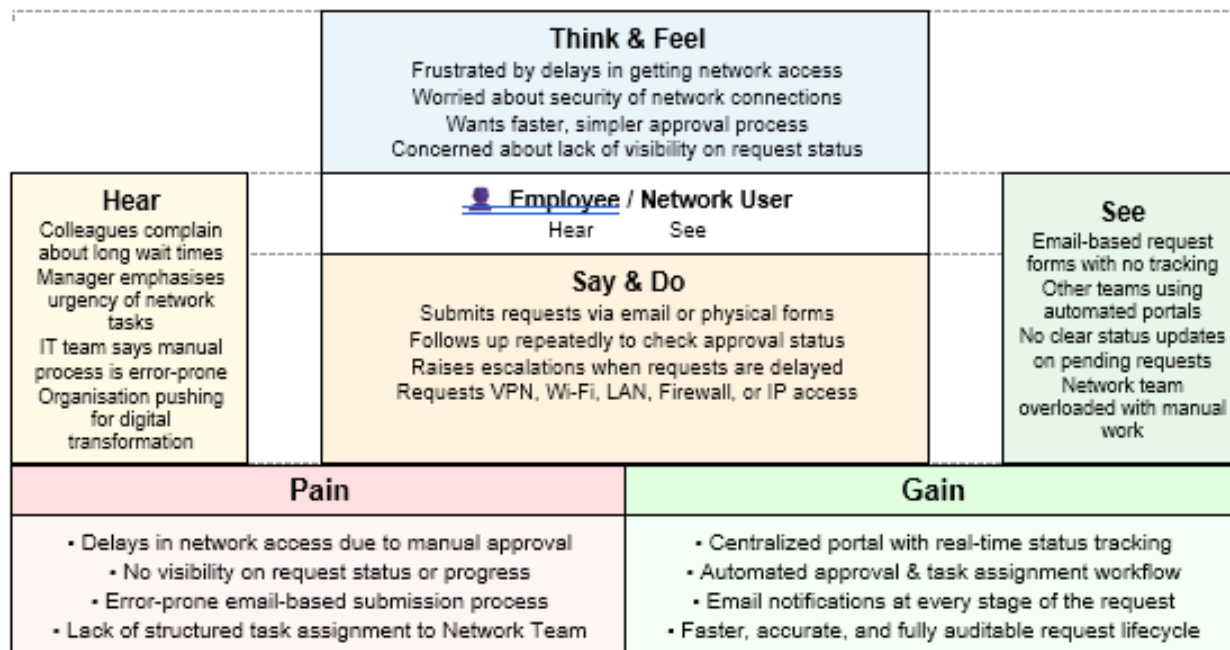
An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to help teams better understand their users.

Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Example:

Network Request Management System



2.3 BRAINSTORM & IDEA PRIORITIZATION

Date	22 June 2026
Team ID	
Project Name	Network Request Management System
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

This template captures the ideation process behind the Network Request Management System — a ServiceNow-based application that automates the handling of network-related service requests (VPN, Wi-Fi, LAN, Firewall, and IP Address access) so employees no longer rely on manual emails or forms.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

The project team gathered to define roles and align on the goal of automating network service requests within the organization. Stakeholders included Employees, Managers, Network Engineers, and System Administrators, each contributing perspective on current pain points with the manual, email-based request process.



Network Request Management System

ServiceNow application for automating network service requests — VPN, Wi-Fi, LAN, Firewall, and IP Address access. Replaces manual emails with a tracked, automated approval workflow.

- Service Catalog item: Network Request
- Roles: Employee, Manager, Network Engineer, System Administrator
- Flow Designer drives approval → task → completion → notification

Problem Statement (How Might We):

How might we replace slow, untracked, email-based network access requests with a centralized, automated, and auditable system that speeds up approvals and improves transparency for both employees and the network team?

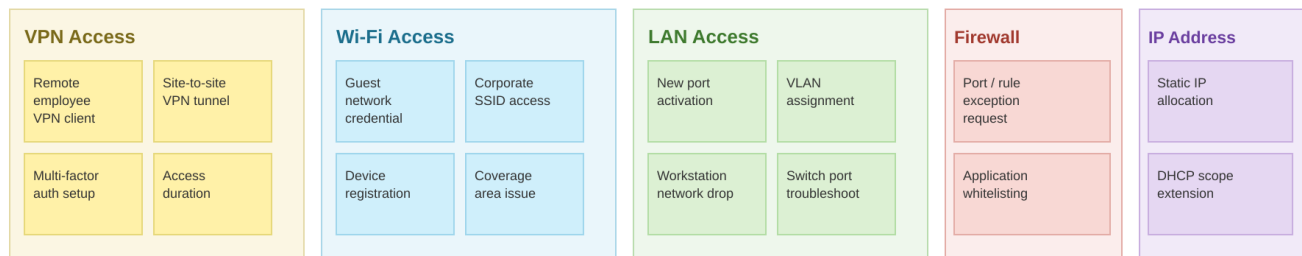
Key rules of brainstorming applied:

- Stay on topic focus on network request handling only
- Encourage wild ideas consider full automation, not partial fixes
- Defer judgment capture every request type before filtering
- Listen to others corporate input from Network Engineers and Admins
- Go for volume, then prioritize list all request types, then rank them

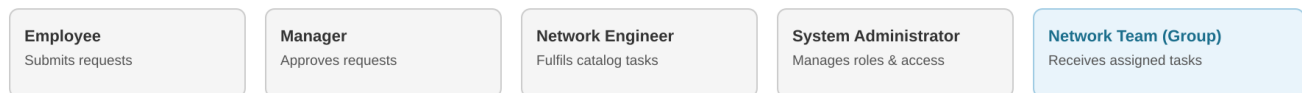
Step-2: Brainstorm, Idea Listing and Grouping

The team brainstormed every network-related service request employees commonly raise, then grouped the resulting ideas into five categories that map directly to the Request Type options on the Network Request Service Catalog item. Supporting roles, the Network Team group, and the custom user_network table used to store request data were also identified during this step.

Network Request Types — Grouped by Category



Roles & Groups Involved



Supporting table: user_network — custom ServiceNow table used to store and manage network request records.

Each colored block represents a request sub-type collected via the Service Catalog item's "Request Type" select box variable.

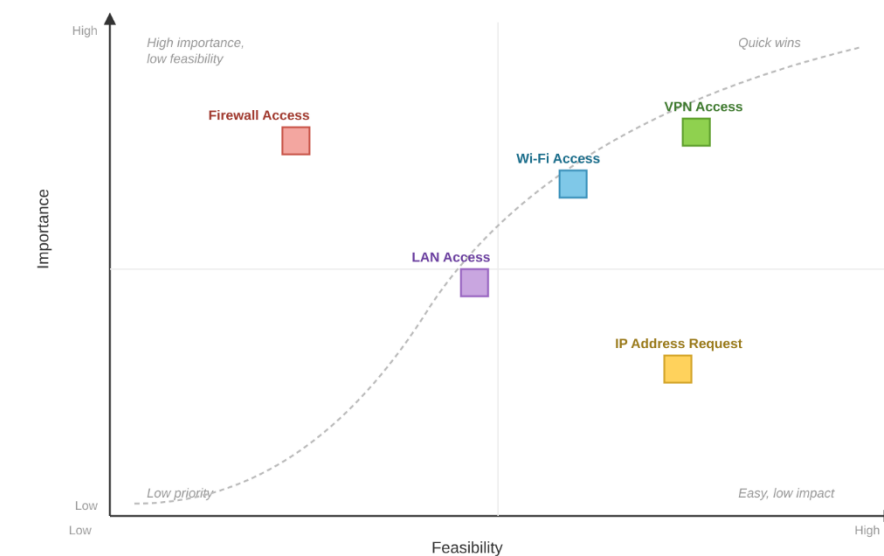
Resulting request categories:

1. **VPN Access** — remote and site-to-site VPN, MFA setup, access duration
2. **Wi-Fi Access** — guest credentials, corporate SSID, device registration, coverage issues
3. **LAN Access** — new port activation, VLAN assignment, network drops, troubleshooting
4. **Firewall Access** — port/rule exceptions, application whitelisting
5. **IP Address Request** — static IP allocation, DHCP scope extension

Step-3: Idea Prioritization

Once grouped, the five request categories were plotted on an Importance vs. Feasibility matrix to decide build order. VPN and Wi-Fi Access scored highest on both axes since they affect the most employees and reuse existing ServiceNow approval and catalog patterns. Firewall Access, while important for security, needs more review and is positioned as a high-importance, lower-feasibility item in the initial release. Firewall Access, while important for security, needs more review and is positioned as a high-importance, lower-feasibility item in the initial release.

Idea Prioritization — Importance vs. Feasibility



VPN and Wi-Fi access are prioritized first (high importance, high feasibility); Firewall access requires more review before approval.

Resulting build priority:

1. VPN Access and Wi-Fi Access — implemented first as quick wins
2. LAN Access — implemented alongside the core catalog item
3. IP Address Request — included as a lower-effort addition
4. Firewall Access — scoped for the approval flow with extra review before task assignment

Outcome

This ideation output directly shaped the build of the Network Request Management System: the five prioritized categories became the options in the Request Type select box variable, the identified roles informed user and group setup, and the Network Team group and user_network table emerged as supporting structures for the automated approval flow that follows in later phases of the project.

3. REQUIREMENT ANALYSIS

3.1 SOLUTION REQUIREMENTS (FUNCTIONAL & NON-FUNCTIONAL)

Date	22 June 2026
Team ID	
Project Name	Network Request Management System
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Network Request Submission	Request VPN Access Request Wi-Fi Access Request LAN Access Request Firewall Access Request IP Address Allocation
FR-2	Dynamic Form Behavior	Show/hide fields based on selected Request Type Mandatory field validation before submission Attachment upload for supporting documents
FR-3	Approval Workflow	Route request to Manager for approval Approve request with comments Reject request with comments and notify requester
FR-4	Task Creation & Assignment	Auto-create Catalog Task on approval Auto-assign task to Network Team group

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
		Update task state to Work in Progress
FR-5	Status Tracking	View real-time request status (Open / WIP / Closed Complete) Update parent request status from task status
FR-6	Notifications	Email notification to requester on completion Email notification to requester on rejection
FR-7	User & Role Management	Create users with roles: Employee, Manager, Network Engineer, Administrator Create and maintain Network Team group
FR-8	Data Management	Store and manage request data in the user_network custom table

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The Service Portal form must be intuitive, with dynamic fields based on Request Type so employees can submit a request in under 2 minutes without training.
NFR-2	Security	Access to requests and tasks is controlled via roles and ACLs; only authorized users (Employee, Manager, Network Engineer, Administrator) can view or act on relevant records. Data is encrypted in transit (TLS) and at rest.
NFR-3	Reliability	The approval flow must reliably trigger task creation and notifications with no manual intervention; failed flow executions should be logged and retried automatically.
NFR-4	Performance	Catalog form and request list views should load within 2-3 seconds under normal load; the Flow Designer workflow should complete approval-to-task creation within a few seconds of approval.
NFR-5	Availability	The application should be available 99.9% of the time, leveraging ServiceNow's cloud-hosted multi-instance availability and redundancy.
NFR-6	Scalability	The system should support a growing number of concurrent requests and users without requiring redesign, leveraging ServiceNow's multi-instance SaaS architecture.
NFR-7	Maintainability	Components (Catalog Client Script, Flow Designer flow, ACLs) should be modular and documented so future Network Engineers or Administrators can extend request types easily.
NFR-8	Auditability	All approvals, status changes, and task updates should be tracked with timestamps and user attribution for compliance and reporting purposes.

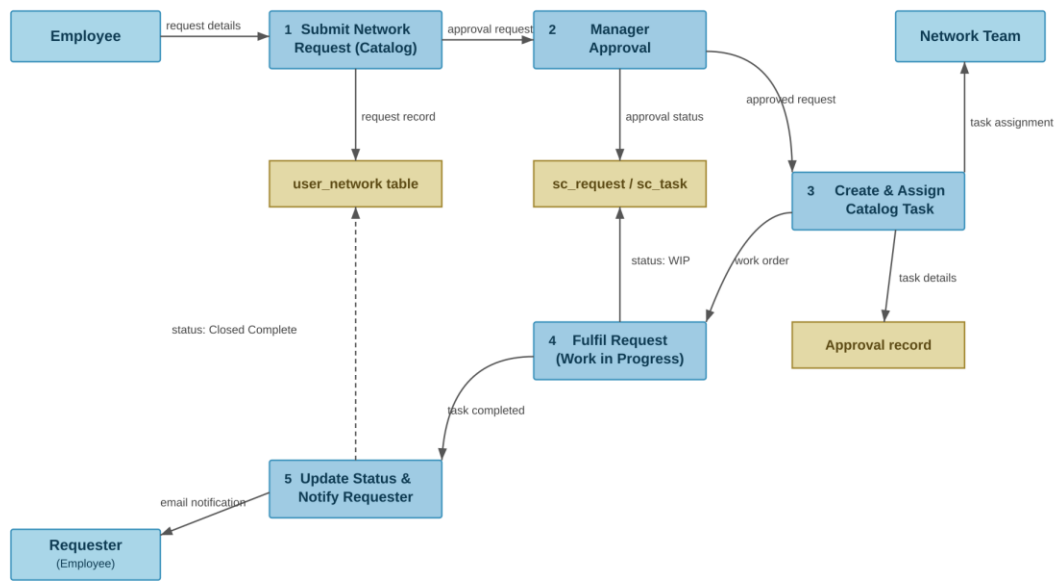
3.2 DATA FLOW DIAGRAM & USER STORIES

Date	22 June 2026
Team ID	
Project Name	Network Request Management System
Maximum Marks	4 Marks

Data Flow Diagram

A Data Flow Diagram (DFD) is a visual representation of how information moves through a system. The diagram below shows how a network access request flows through the Network Request Management System — from the employee submitting the request in the Service Catalog, through manager approval, to task creation, fulfilment by the Network Team, and final status update with email notification back to the requester.

DFD Level-0: Network Request Management System



DFD Level-0: Network Request Management System — from submission to completion notification.

Flow summary:

1. Employee submits a Network Request through the Service Catalog (request type, attachment, and supporting details are captured and written to the user_network table).
2. The Network Request Approval Flow routes the request to the appropriate Manager for approval.
3. On approval, the flow automatically creates a Catalog Task and assigns it to the Network Team group.
4. The Network Engineer updates the task to Work in Progress while fulfilling the request.
5. Once complete, the task and request status are updated to Closed Complete, and an email notification is sent to the requester.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Employee	Submit Request	USN-1	As an employee, I can select "Network Request" from the Service Catalog so I can raise a request for VPN, Wi-Fi, LAN, Firewall, or IP access.	I can open the Network Request catalog item and see all request type options.	High	Sprint-1
		USN-2	As an employee, I can choose a request type from a select box so that only the relevant fields are shown to me.	Selecting a request type dynamically shows/hides the matching form fields via client script.	High	Sprint-1
		USN-3	As an employee, I can attach supporting documents to my request so the approver has the context needed to decide.	I can upload an attachment and it is visible on the submitted request.	Medium	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
	Track Request	USN-4	As an employee, I can view the status of my submitted request so I know whether it is pending, in progress, or completed.	I can see real-time status (Open / Work in Progress / Closed Complete) on my request.	High	Sprint-2
	Notifications	USN-5	As an employee, I can receive an email notification once my request is completed so I don't need to follow up manually.	I receive an email when the request status changes to Closed Complete.	High	Sprint-1
Manager	Approve Request	USN-6	As a manager, I can review a submitted network request so I can decide whether to approve or reject it.	I can see requester details, request type, and justification before approving.	High	Sprint-1
		USN-7	As a manager, I can approve or reject a request with comments so the requester understands the decision.	Approving triggers the flow to create a catalog task; rejecting notifies the requester with my comment.	High	Sprint-1
Network Engineer	Fulfil Request	USN-8	As a network engineer, I can view catalog tasks assigned to the Network Team so I know which requests need action.	Approved requests appear as tasks assigned to the Network Team group.	High	Sprint-2
		USN-9	As a network engineer, I can update a task to Work in Progress so the requester and manager can track active work.	Updating task state to Work in Progress reflects on the parent request status.	Medium	Sprint-2
		USN-10	As a network engineer, I can mark a task as Closed	Closing the task updates the	High	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
			Complete once the access is granted so the request lifecycle finishes correctly.	request status and triggers the completion email.		
System Administrator	User & Role Management	USN-11	As a system administrator, I can create users and assign roles (Employee, Manager, Network Engineer, Administrator) so access is controlled correctly.	Each role has the correct permissions on the Network Request catalog item and tasks.	High	Sprint-1
	Group Management	USN-12	As a system administrator, I can create and maintain the Network Team group so approved tasks route to the right people.	Catalog tasks are auto-assigned to the Network Team group on approval.	Medium	Sprint-1
	Table Management	USN-13	As a system administrator, I can manage the user_network custom table so network request data is stored and retrievable when needed.	Request records are correctly written to and readable from the user_network table.	Medium	Sprint-2
	Workflow Configuration	USN-14	As a system administrator, I can configure and maintain the Network Request Approval Flow so the approval-to-completion process runs without manual intervention.	The Flow Designer workflow triggers approval, task creation, status updates, and email notification automatically.	High	Sprint-2

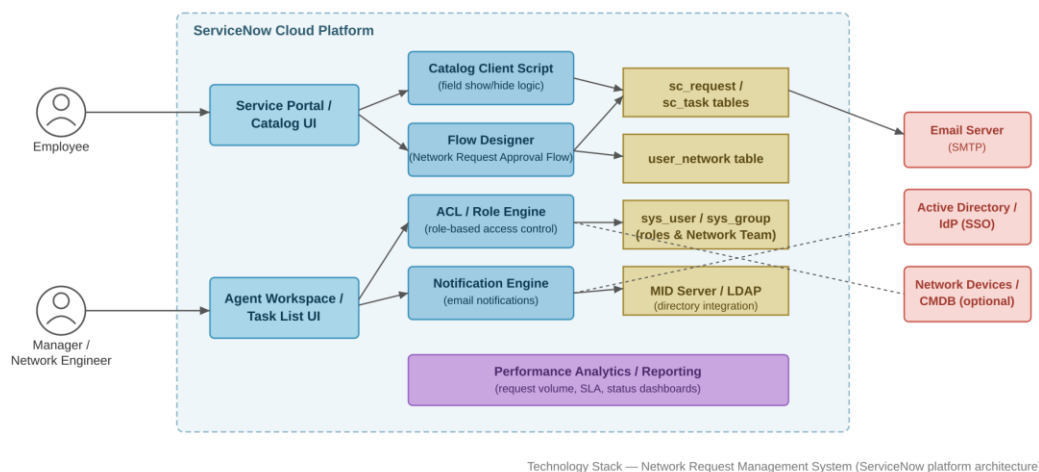
3.3 TECHNOLOGY STACK (ARCHITECTURE & STACK)

Date	22 June 2026
Team ID	
Project Name	Network Request Management System
Maximum Marks	4 Marks

Technical Architecture

The deliverable includes the architectural diagram below along with the component and characteristic details captured in Table-1 and Table-2.

Example: Network Request Management System on ServiceNow



Guidelines applied:

- All processes included as application-logic blocks: Catalog Client Script, Flow Designer, ACL/Role Engine, and Notification Engine
- Infrastructural demarcation shown via the ServiceNow Cloud Platform boundary separating internal logic from external systems
- External interfaces indicated: Email Server (SMTP), Active Directory / IdP (SSO), and optional Network Devices / CMDB integration
- Data storage components indicated: sc_request/sc_task, user_network, sys_user/sys_group, and MID Server/LDAP
- No machine learning model is used in the current scope of this application

Table-1: Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How the user interacts with the application — Service Portal, mobile app access to catalog item.	ServiceNow Service Portal (Now Experience UI), Mobile App (Now Mobile)
2.	Application Logic-1	Logic to dynamically show/hide form fields based on selected Request Type.	Catalog Client Script (JavaScript, glideform API)
3.	Application Logic-2	Workflow logic for routing approvals, creating tasks, updating status, and triggering notifications.	Flow Designer (Network Request Approval Flow)
4.	Application Logic-3	Logic for controlling who can view, submit, or act on requests based on assigned role.	Access Control Lists (ACLs), Roles & Groups
5.	Database	Data type, configuration, and storage of request records, tasks, and custom data.	ServiceNow native tables (sc_request, sc_task, user_network) on MySQL-based instance database
6.	Cloud Database	Database service hosted on cloud.	ServiceNow Instance Database (multi-instance cloud architecture)
7.	File Storage	Storage for attachments uploaded with a request (e.g. approval documents, screenshots).	ServiceNow Attachment (sys_attachment) table / native ServiceNow file storage
8.	External API-1	Purpose of external API used in the application.	Email/SMTP integration for completion notifications
9.	External API-2	Purpose of external API used in the application.	LDAP / Active Directory or SSO integration for user authentication (optional)
10.	Machine Learning Model	Purpose of Machine Learning model (not used in current scope).	Not applicable in current release
11.	Infrastructure (Server / Cloud)	Application deployment on local system / cloud. Cloud server configuration.	ServiceNow multi-instance SaaS (Cloud) — Dev, Test, and Production instances

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used.	AngularJS (used internally by ServiceNow Service Portal widgets), jQuery
2.	Security Implementations	List all the security / access controls implemented, use of firewalls, etc.	Role-Based Access Control (RBAC), Access Control Lists (ACL), Data encryption at rest and in transit (TLS), MFA / SSO for login
3.	Scalable Architecture	Justify the scalability of architecture (multi-instance, cloud-hosted SaaS model).	ServiceNow multi-instance cloud architecture; horizontally scalable across customer instances
4.	Availability	Justify the availability of application — load balancing and redundancy handled by the platform.	ServiceNow Cloud (built-in high availability, load balancers, redundant data centers)
5.	Performance	Design consideration for the performance of the application — number of requests per sec, caching, etc.	Performance Analytics dashboards, indexed tables, scheduled jobs for SLA tracking

4. PROJECT DESIGN

4.1 PROBLEM – SOLUTION FIT

Date	22 June 2026
Team ID	
Project Name	Network Request Management System
Maximum Marks	2 Marks

 PROBLEM Employees submit network service requests (VPN, Wi-Fi, LAN, Firewall, IP Address) through manual emails with no standardized process, causing delays, no visibility, and improper task assignment to the Network Team.	 EXISTING ALTERNATIVES Sending emails to IT/Network Team manually. Following up via phone calls. Raising verbal requests to managers. No tracking, no structured approval, and no automated task handoff.	 KEY FEATURES • Service Catalog item with 5 request types • Automated approval via Flow Designer • Auto-created Catalog Task for Network Team • Email notifications at every stage • Catalog Client Script for dynamic fields • Custom user_network table for record management	 EARLY ADOPTERS Employees who frequently request network access. New joiners needing VPN or Wi-Fi setup. Managers who approve network requests. Network Engineers assigned to fulfill access tasks.	 SOLUTION Centralized Network Request Management System on ServiceNow — Service Catalog item with automated Flow Designer approval, role-based access control, auto Catalog Task creation for the Network Team, and email notifications on request completion.
 CUSTOMER SEGMENT Employees (Requestors) in an organization who need VPN, Wi-Fi, LAN, Firewall, or IP Address access for their daily work. Managers who authorize network access requests. Network Engineers who fulfill and close approved tasks.	 UNIQUE VALUE PROPOSITION One-click network service request submission with automated approval routing, real-time status tracking, and structured task assignment — eliminating all manual email-based IT coordination.	 UNFAIR ADVANTAGE Built entirely on the ServiceNow platform using native tools — Service Catalog, Flow Designer, Role Management, and Custom Tables. No external integration needed. The workflow is fully reusable and scalable to any network service type or department.		

4.2 PROPOSED SOLUTION

Date	22 June 2026
Team ID	
Project Name	Network Request Management System
Maximum Marks	2 Marks

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Employees in an organization request network access (VPN, Wi-Fi, LAN, Firewall, IP Address) through manual emails and phone calls, causing delays, lack of visibility, compliance issues with network security policies, and increased workload for the network team with no standardized process.
2.	Idea / Solution description	A ServiceNow-based Network Request Management System using a standardized Service Catalog item, Flow Designer workflow for automated approvals, auto-created fulfilment tasks for the Network Team, role-based access control for exception handling, and email notifications at every stage all accessible through the ServiceNow Service Portal.
3.	Novelty / Uniqueness	Unlike manual email-based processes, this solution provides a fully automated end-to-end request lifecycle within ServiceNow from submission to task fulfilment with zero manual coordination, real-time tracking, and built-in role-based access enforcement through Access Control Lists.
4.	Social Impact / Customer Satisfaction	Employees get faster network access with full visibility into request status. Network Engineers spend less time on manual coordination and more time on critical fulfilment tasks. Managers get structured, policy-compliant approval requests. Overall IT service satisfaction improves significantly.
5.	Business Model (Revenue Model)	This is an internal IT automation project for an organization. Cost savings are achieved by reducing network/IT support hours spent on manual request handling, minimizing unauthorized or untracked access changes, and ensuring policy compliance directly reducing operational costs.
6.	Scalability of the Solution	The solution is built on ServiceNow's native Flow Designer and Service Catalog making it easily scalable to handle additional request types, departments, and approval groups. Additional catalog items, workflow branches, and notification rules can be added without redesigning the existing system.

4.3 SOLUTION ARCHITECTURE

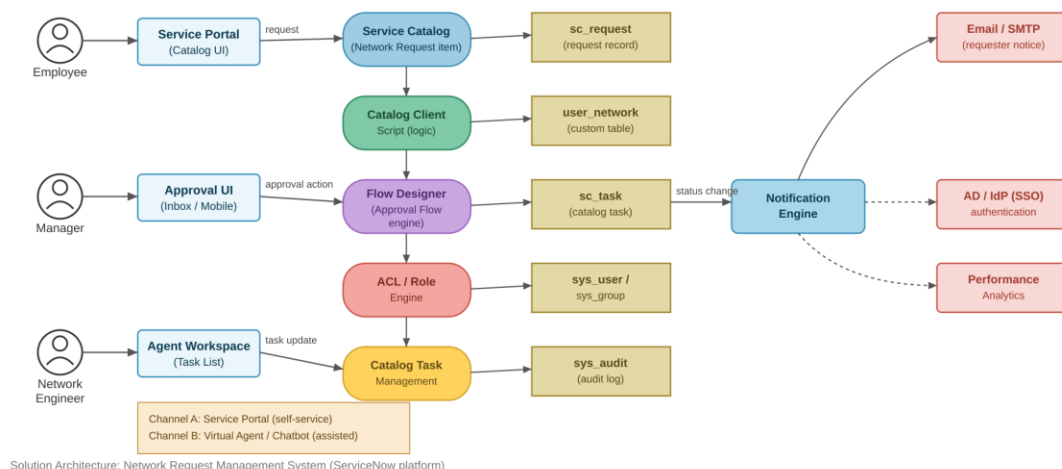
Date	22 June 2026
Team ID	
Project Name	Network Request Management System
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process — with many sub-processes — that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram -Network Request Management System:



5. PROJECT PLANNING AND SCHEDULING

5.1 PROJECT PLANNING

Date	22 June 2026
Team ID	
Project Name	Network Request Management System
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Network Request Submission	USN-1	As an employee, I can select “Network Request” from the Service Catalog and choose VPN, Wi-Fi, LAN, Firewall, or IP Address as my request type.	3	High	NITHISHA B
Sprint-1		USN-2	As an employee, I can see only the fields relevant to my selected request type, since irrelevant fields are hidden automatically.	2	High	NITHISHA B
Sprint-1		USN-3	As an employee, I can attach supporting documents to my request so the approver has the context needed to decide.	1	Medium	NITHISHA B
Sprint-2	Approval Workflow	USN-4	As a manager, I can review a submitted network request with requester details and justification before deciding.	2	High	NITHISHA B
Sprint-2		USN-5	As a manager, I can approve or reject a request with comments so the requester understands the decision.	3	High	NITHISHA B
Sprint-2	Task Creation & Assignment	USN-6	As the system, I can auto-create a Catalog Task and assign it to the Network Team group once a request is approved.	3	High	NITHISHA B

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-2		USN-7	As a network engineer, I can view catalog tasks assigned to the Network Team so I know which requests need action.	1	Medium	NITHISHA B
Sprint-3	Fulfilment & Status Tracking	USN-8	As a network engineer, I can update a task to Work in Progress so the requester and manager can track active work.	2	Medium	NITHISHA B
Sprint-3		USN-9	As a network engineer, I can mark a task as Closed Complete once access is granted so the request lifecycle finishes correctly.	2	High	NITHISHA B
Sprint-3		USN-10	As an employee, I can view the real-time status of my submitted request (Open / WIP / Closed Complete).	2	High	NITHISHA B
Sprint-3	Notifications	USN-11	As an employee, I can receive an email notification once my request is completed so I don't need to follow up manually.	1	High	NITHISHA B
Sprint-1	User & Role Management	USN-12	As a system administrator, I can create users and assign roles (Employee, Manager, Network Engineer, Administrator) so access is controlled correctly.	3	High	NITHISHA B
Sprint-1	Group Management	USN-13	As a system administrator, I can create and maintain the Network Team group so approved tasks route to the right people.	1	Medium	NITHISHA B
Sprint-2	Data Management	USN-14	As a system administrator, I can manage the user_network custom table so network request data is stored and retrievable when needed.	2	Medium	NITHISHA B

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	10	5 Days	20 June 2026	24 June 2026	10	24 June 2026
Sprint-2	12	5 Days	25 June 2026	29 June 2026	12	29 June 2026

Velocity:

Total Story Points = Sprint 1 + Sprint 2 = 10 + 12 = 22

Number of Sprints = 2

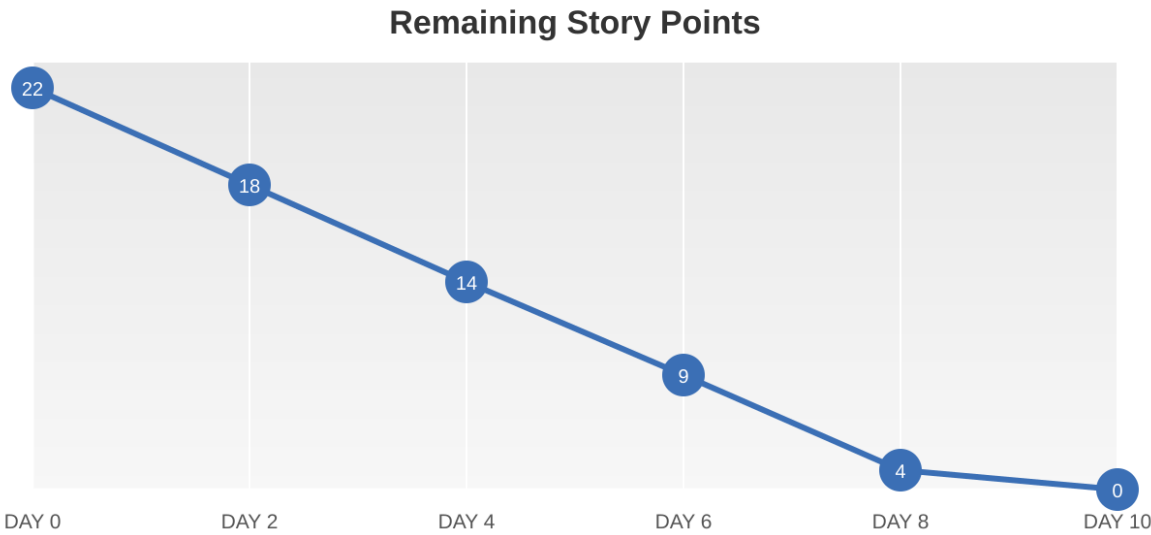
Velocity = Total Story Points ÷ Number of Sprints = 22 ÷ 2 = 11 Story Points per Sprint

The team's velocity is 11 Story Points per Sprint.

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over tim

e.



6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 USER ACCEPTANCE TESTING (UAT)

Date	22 June 2026
Team ID	
Project Name	Network Request Management System
Maximum Marks	4 Marks

Project Overview:

Project Name: Network Request Management System

Project Description: A ServiceNow-based application that automates the submission, approval, fulfilment, and tracking of employee network access requests (VPN, Wi-Fi, LAN, Firewall, and IP Address access) through a Service Catalog item and an automated Flow Designer approval workflow.

Project Version: 1.0

Testing Period: 20 June 2026 to 28 June 2026

Testing Scope:

Features and functionalities to be tested:

- Network Request catalog item, including dynamic field display by Request Type
- Manager approval and rejection workflow (Network Request Approval Flow)
- Catalog Task auto-creation and assignment to the Network Team
- Task fulfilment and status updates (Work in Progress, Closed Complete)
- Email notifications to the requester on completion or rejection

User stories / requirements to be tested: USN-1 through USN-14 (see Solution Requirements document).

Testing Environment:

URL/Location: <https://dev391575.servicenow.com>

Credentials (if required): 927623bad069@mkce.ac.in & Nithi@003

Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	Submit a VPN access request	1. Log in as Employee 2. Open Network Request catalog item 3. Select Request Type = VPN Access 4. Fill required fields and submit	Request is created with status Open and the correct fields for VPN Access are shown	As expected	Pass
TC-002	Dynamic field visibility on Request Type change	1. Open Network Request catalog item 2. Switch Request Type between VPN, Wi-Fi, LAN, Firewall, IP	Only fields relevant to the selected request type are shown; others are hidden automatically	As expected	Pass
TC-003	Attach supporting document to request	1. Open Network Request catalog item 2. Attach a file 3. Submit the request	Attachment is saved and visible on the submitted request record	As expected	Pass
TC-004	Manager approves a request	1. Log in as Manager 2. Open pending approval 3. Click Approve with a comment	Request status updates to Approved; a Catalog Task is auto-created and assigned to the Network Team	As expected	Pass
TC-005	Manager rejects a request	1. Log in as Manager 2. Open pending approval 3. Click Reject with a comment	Request status updates to Rejected; requester receives an email with the rejection comment	As expected	Pass
TC-006	Network Engineer fulfils a task	1. Log in as Network Engineer 2. Open assigned Catalog Task 3. Update state to Work in Progress 4. Close task as Complete	Task and parent request status update to Closed Complete; requester receives a completion email	As expected	Pass
TC-007	Unauthorized user attempts to approve a request	1. Log in as Employee (non-manager) 2. Attempt to open an approval record directly via URL	Access is denied due to ACL restrictions	As expected	Pass

Bug Tracking:

Bug ID	Bug Description	Steps to reproduce	Severity	Status	Additional feedback
BG-001	Firewall Access fields remain visible after switching to VPN Access request type	1. Open Network Request catalog item 2. Select Request Type = Firewall Access 3. Switch Request Type to VPN Access	Medium	Open	Client script condition for Firewall-specific fields needs an additional onChange check
BG-002	Approval email not sent to Manager when request is submitted after business hours	1. Submit a request after 6 PM 2. Check Manager's inbox	High	In Progress	Notification trigger appears to depend on a scheduled job; recommend switching to event-driven trigger
BG-003	Catalog Task not auto-assigned to Network Team for IP Address Request type	1. Submit and approve an IP Address Request 2. Check Catalog Task assignment group	High	Closed	Fixed by correcting the assignment group reference in Flow Designer for this request type

Sign-off:

Tester Name: NITHISHA B

Date: 28 June 2026

Signature: NITHISHA B

Notes:

- Ensure that all test cases cover both positive and negative scenarios (e.g. valid approvals and unauthorized access attempts).
- Encourage testers to provide detailed feedback, including any suggestions for improvement.
- Bug tracking should include details such as severity, status, and steps to reproduce.
- Obtain sign-off from both the project manager and product owner (or equivalent Network Team lead) before proceeding with deployment.

6.2 ACCEPTANCE TESTING UAT

Execution & Report Submission

Date	22 June 2026
Team ID	
Project Name	Network Request Management System
Maximum Marks	4 Marks

1. Purpose of Document

The purpose of this document is to briefly explain the test coverage and open issues of the Network Request Management System project at the time of release to User Acceptance Testing (UAT).

2. Defect Analysis

This report shows the number of resolved or closed bugs at each severity level, and how they were resolved.

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	4	2	1	1	8
Duplicate	1	0	1	0	2
External	0	1	0	1	2
Fixed	6	3	2	9	20
Not Reproduced	0	0	1	0	1
Skipped	0	0	1	1	2
Won't Fix	0	2	1	0	3
Totals	11	8	7	12	38

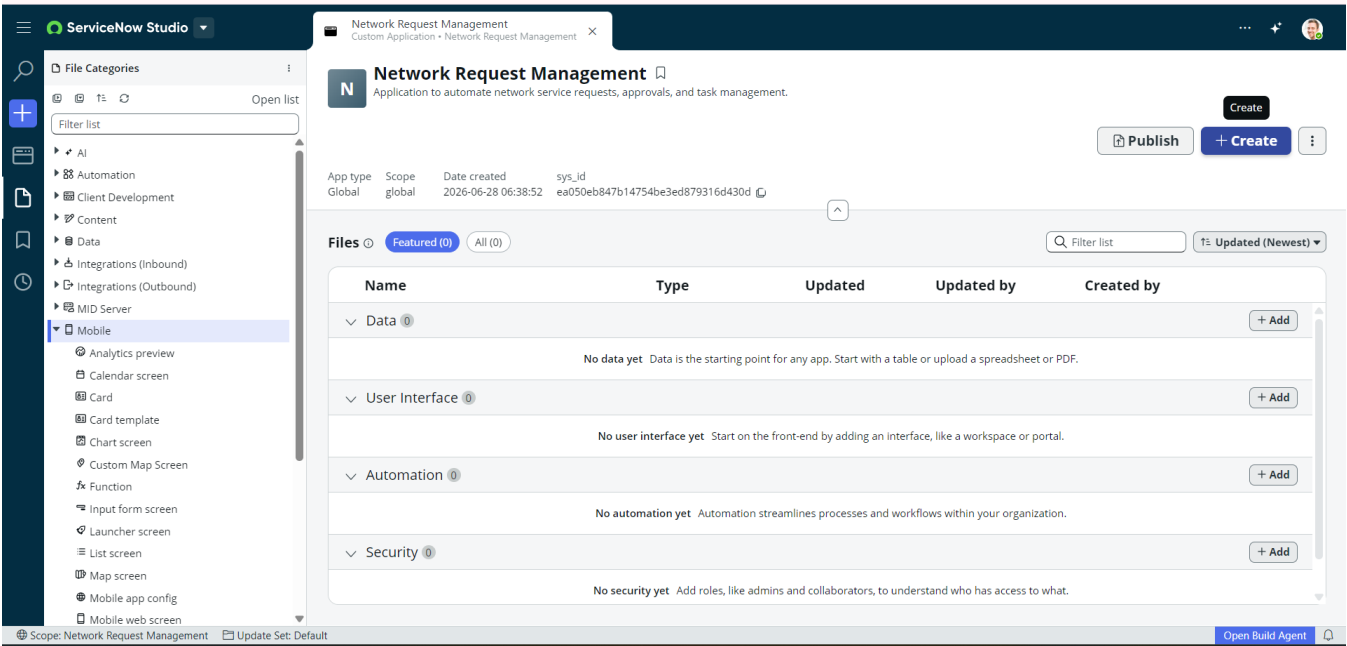
3. Test Case Analysis

Section	Total Cases	Not Tested	Fail	Pass
Network Request Catalog Form	14	0	1	13
Approval Workflow (Flow Designer)	10	0	0	10
Catalog Task Management	9	0	1	8
Roles & Access Control (ACL)	6	0	0	6
Email Notifications	5	0	1	4
user_network Custom Table	4	1	0	3
Reporting & Performance Analytics	3	0	0	3
Totals	51	1	3	47

This report shows the number of test cases that have passed, failed, and untested.

7. RESULTS

7.1 OUTPUT SCREENSHOTS



ServiceNow Studio

Network Request Management
Custom Application • Network Request Management

user_network - Default view
Table • Network Request Management

user_network

DataFormsFlows

Policies and rules

Network Reque...

PreviewSave

File Categories

Open list

Filter list

AI

Automation

Client Development

Content

Data

Integrations (Inbound)

Integrations (Outbound)

MID Server

Mobile

Analytics preview

Calendar screen

Card

Card template

Chart screen

Custom Map Screen

Function

Input form screen

Launcher screen

List screen

Map screen

Mobile app config

Mobile app config screen

Table fields

Add new field

Filter fields

Filter

Column label *	Column name *	Type *	Reference	Max length	Default value	Display
Number	u_number	String		40	javascript:global.ge...	☐
Priority	u_priority	Choice 3 Choices				☐
Request Type	u_request_type	Choice 4 Choices				☐
Resolution Notes	u_resolution_notes	String		40		☐
Status	u_status	Choice 6 Choices				☐
Updated	sys_updated_on	Date/Time				☐
Updated by	sys_updated_by	String		40		☐
Updates	sys_mod_count	Integer				☐
Work Notes	u_work_notes	Journal Input				☐

Add new field

Showing 1-17 of 17

1

Records per page

servicenow

AllFavoritesHistoryWorkspacesAdmin

Groups

Search

Search

Actions on selected rows...

New

All > Name starts with network

Name	Description	Active	Manager	Parent	Updated
network	Search	Search	Search	Search	Search
Network		true	Bow Ruggeri	(empty)	2023-06-06 19:02:08
Network Admin	Full control of the application	true	(empty)	(empty)	2026-06-28 00:20:58
Network Approvers	Approve network requests	true	(empty)	(empty)	2026-06-28 00:19:36
Network CAB Managers	Group to represent the CAB Managers for ...	true	(empty)	(empty)	2026-03-31 01:36:33
Network Requesters	Employees submit requests	true	(empty)	(empty)	2026-06-28 00:20:27
Network Team	Engineers work on network tasks	true	(empty)	(empty)	2026-06-28 00:18:50

1 to 6 of 6

dev391575.service-now.com/now/nav/ui/classic/params/target/sys_user_role_list.do%3Fsysparm_query%3DnameSTARTSWITHnetwork%26sysparm_first_row%3D1%26sysp...

servicenow

AllFavoritesHistoryWorkspacesAdmin

Roles

Search

Search

Actions on selected rows...

New

All > Name starts with network

Name	Description	Elevated privilege
network	Search	Search
network_admin	Full access to the application	false
network_approver	Manager can approve/reject requests	false
network_engineer	Network team can work on tasks	false
network_requester	Employee can submit network requests	false
network_user		false

1 to 5 of 5

dev391575.service-now.com/now/nav/ui/classic/params/target/sys_user_list.do%3Fsysparm_query%3Duser_nameSTARTSWITHemployee1%26sysparm_first_row%3D1...

servicenow

AllFavoritesHistoryWorkspacesAdmin

Users

Search

Search

Actions on selected rows...

New

All > User ID starts with employee1

User ID	Name	Email	Active	Created	Updated
employee1	Search	Search	Search	Search	Search
employee1	Employee One	balu40709@gmail.com	true	2026-06-28 00:37:56	2026-06-28 00:37:56

1 to 1 of 1

servicenow

AllFavoritesHistoryWorkspacesAdmin

Users

Search

Actions on selected rows...New

All > User ID starts with manager1

User ID	Name	Email	Active	Created	Updated
manager1	Search	Search	Search	Search	Search
Manager1	Manager One	balu40709@gmail.com	true	2026-06-28 00:43:56	2026-06-28 00:43:56

1 to 1 of 1

dev391575.service-now.com/now/nav/ui/classic/params/target/sys_user_list.do%3Fsysparm_query%3DUser_nameSTARTSWITHEngineer1%26sysparm_first_row%3D1%26sysparm_max_row%3D10

servicenow

AllFavoritesHistoryWorkspacesAdmin

Users

Search

Actions on selected rows...New

All > User ID starts with Engineer1

User ID	Name	Email	Active	Created	Updated
Engineer1	Search	Search	Search	Search	Search
engineer1	Engineer One	balu40709@gmail.com	true	2026-06-28 00:45:37	2026-06-28 00:45:37

1 to 1 of 1

dev391575.service-now.com/now/nav/ui/classic/params/target/sys_user_list.do%3Fsysparm_query%3Duser_nameSTARTSWITHadmin%26sysparm_first_row%3D1%26sysparm...

servicenow

AllFavoritesHistoryWorkspacesAdminUsers

Search

Actions on selected rows...New

All > User ID starts with admin1

User ID	Name	Email	Active	Created	Updated
admin1	Admin One	balu40709@gmail.com	true	2026-06-28 00:46:45	2026-06-28 00:46:45

Catalog Items

NameSearch

Actions on selected rows...New

All > Type != Bundle > Class != Order guide > Type != Package > Class != Content Item > Published item is empty > Name starts with Network Request

Name	Short description	Active	Roles	Catalogs	Category	Price	Type	Updated
Network Request	Request for network access or configuration	true		Service Catalog	Network Standard Changes	\$0.00	Item	2026-06-28 02:08:17

ActivateDeactivate



Workflow Studio | Network Request Approval Flow | Network Request Management

Network Request Approval Flow

TRIGGER

- Service Catalog

ACTIONS

- Ask For Approval on Requested Item
- Create Catalog Task Record
- Update Requested Item Record
- Send Email

ERROR HANDLER

Status: Published | Application: Network Request Management

Data

- Flow Variables
 - Trigger - Service Catalog
 - Requested Item Record (Record)
 - Run Start Time UTC (Date/Time)
 - Table Name (Table Name)
 - Run Start Date/Time (Date/Time)
 - 1 - Ask For Approval
 - Approval State (Choice)
 - Action Status (Object)
 - 2 - Create Record
 - Catalog Task Record (Record)
 - Catalog Task Table (Table)
 - Action Status (Object)
 - 3 - Update Record

How can we help?

Request Something | Knowledge Base | Get Help

Current Status

No system is reporting an issue

More information...

My Assessments and Surveys

No assessments or surveys for you at the moment

My Approvals

You have no pending approvals

Announcements

Employee Center is available to you

My Open Incidents

Email server is down.
INC0009005 • about an hour ago

Top Rated Articles

dev391575.service-now.com/sp?id=sc_cat_item&sys_id=9995567c47354754be3ed879316d4377

☆ 0

servicenow

KnowledgeCatalogRequestsSystem StatusCartToursSystem Administrator

Home > Service Catalog > Standard Changes > Network Standard Changes > Network Request

Search Catalog

Network Request

Request for network access or configuration

Request Type

Attachment

Upload

Add attachments

Choose a file or drag it here.

Copy and paste clipboard files here.

Quantity: 1

Delivery Time: 2 Days

Add to Cart

Save as Draft

Order Now

dev391575.service-now.com/sp?id=sc_cat_item&sys_id=9995567c47354754be3ed879316d4377&sysparm_category=abbcbbbf47700200e90d87e8dee49041

☆ 0

servicenow

firewall.png AttachedSystem Administrator

Home > Service Catalog > Standard Changes > Network Standard Changes > Network Request

Search Catalog

Network Request

Request for network access or configuration

Request Type

Attachment

UploadDelete

Add attachments

Choose a file or drag it here.

Copy and paste clipboard files here.

firewall.png (17.2 KB)

just now

Quantity: 2

Delivery Time: 2 Days

Add to Cart

Save as Draft

Order Now

dev391575.service-now.com/sp?id=sc_request&is_new_order&table=sc_request&sys_id=6f221b74473d4754be3ed879316d43c4

System Administrator

Request Summary - REQ0010001

Submitted : 2026-06-28 05:15:56
Request Number : REQ0010001
Requested for : Admin One
Estimated Delivery : 2026-06-30

Item	Delivery Date	Stage	Price (each)	Quantity	Total
Network Request	2026-06-30	Request Approval	---	2	---

Total: \$0.00

Approval - Student: STU0001019

Search

Approval Student: STU0001019

Update Delete

Approver System Administrator

Approving Student: STU0001019

State Rejected

Approval Reason

Comments

Post

Activities: 4

System Administrator

no need

Comments • 2026-06-28 05:29:24

System Administrator

State Rejected was Approved

Field changes • 2026-06-28 05:29:24

System Administrator

State Approved was Requested

Field changes • 2026-06-01 21:53:29

System Administrator

Approver System Administrator
State Requested

Field changes • 2026-06-01 21:50:34

Update Delete

Created	Recipients	Subject	State
2026-06-28 01:00:15	aileen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	Ready
2026-06-28 05:16:13	admin@example.com, balu40709@gmail.com	Request REQ0010001 was approved	Ready
2026-06-28 01:00:15	aileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	Ready
2026-06-28 01:00:13	aileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	Ready
2026-06-28 00:00:13	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	Ready
2026-06-28 05:16:03	fred.luddy@example.com	Catalog Task SCTASK0010002 has been assigned to group Software	Ready
2026-06-28 05:16:03	admin@example.com, balu40709@gmail.com	Request REQ0010001 was created	Ready

8. ADVANTAGES AND DISADVANTAGES

Advantages

The Network Request Management System offers several advantages by automating the complete network request process. It reduces manual paperwork and minimizes the use of emails for requesting network services. The system provides a centralized platform where employees can easily submit requests through the Service Catalog. Automated approval workflows reduce processing time and improve operational efficiency. Tasks are automatically assigned to the appropriate network team, eliminating manual task allocation. Users can track the status of their requests in real time, which improves transparency and communication. Role-based access control enhances system security by allowing only authorized users to perform specific actions. Email notifications keep requesters informed throughout the request lifecycle, ensuring better user satisfaction. The system also maintains a complete history of requests, making auditing and reporting easier.

Disadvantages

The system depends entirely on the ServiceNow platform and requires proper licensing and internet connectivity. Initial configuration and workflow development require technical knowledge of ServiceNow administration and development. If workflows or approvals are configured incorrectly, requests may not be processed as expected. Organizations may also need to train employees before they can effectively use the system. Integration with external network management tools may require additional customization and development effort.

9. CONCLUSION

The Network Request Management System successfully automates the complete lifecycle of network service requests using the ServiceNow platform. Employees can easily submit network requests through the Service Catalog, while managers approve or reject requests through an automated approval process. Once approved, the system automatically creates tasks for the Network Team, updates request statuses, and sends email notifications to users. The project improves efficiency, reduces manual effort, enhances transparency, and provides better tracking of network requests. Overall, the system demonstrates the practical implementation of ServiceNow Administration, Service Catalog, Flow Designer, Client Scripts, Workflow Automation, and Notification Management.

10. FUTURE SCOPE

The project can be further enhanced by implementing multiple approval levels based on request type or business criticality. Service Level Agreements (SLAs) can be added to monitor request completion time and improve service quality. The application can be integrated with Active Directory for automatic user account provisioning and network access management. Additional communication methods such as Microsoft Teams, Slack, or SMS notifications can be integrated. AI-powered recommendations can assist users in selecting appropriate request types and predicting approval outcomes. Advanced dashboards and analytics can be developed to monitor request trends, team performance, and SLA compliance. Mobile support can also be enhanced to allow users to submit and track requests through the ServiceNow mobile application.

11. APPENDIX

Modules Developed:

- User Management
- Role Management
- Group Management
- Service Catalog
- Catalog Variables
- Catalog Client Script
- Flow Designer

- Approval Workflow
- Catalog Task Creation
- Email Notifications

Tools Used:

- ServiceNow Developer Instance
- Flow Designer
- Service Catalog
- JavaScript (Catalog Client Script)

Source Code:

```
function onChange(control, oldValue, newValue, isLoading) {  
  
    if (isLoading || newValue === '') {  
        return;  
    }  
  
    if (newValue === 'vpn') {  
        g_form.setDisplay('access_level', true);  
    } else {  
        g_form.setDisplay('access_level', false);  
    }  
}
```

GitHub Link :

<https://github.com/Nithishabalu/Network-Request-Management-ServiceNow>

Project Demo Link :

<https://drive.google.com/file/d/1BDcOHX9OcmqcrNsdotp640TLg18EUNy0/view?usp=sharing>